

Retribution and the Goals of Punishment

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1. Study information

1.1 Background

Criminal punishment is usually defended by one of four goals: retribution, deterrence, incapacitation, and rehabilitation. An earlier study in this line of work, *the Dark Side of Punishment*, found that people who favor more punishment also tend to endorse attitudes that none of these four goals can justify, among them hatred of offenders, a wish for revenge, support for degrading prisoners, tolerance of prison violence, and a willingness to weaken the protections that defendants receive.

This study has two aims. The first is to repeat the Dark Side result in a new and larger sample. The second is to look more closely at retribution. Retribution is the goal people invoke most often, and it carries the most demanding conditions. A defensible retributive position holds that punishment must be deserved, must be proportionate to the offense, and must respect several limits that Kant set out: guilt has to be established through fair procedure and reliable evidence, punishment should not be driven by hatred, and it should not strip the offender of dignity or impose needless suffering. We ask whether ordinary support for retribution meets these conditions, or whether what people call just deserts is closer to revenge.

We single out participants who rank retribution first among the four goals. We call them retributivists, compare them with everyone else, and ask whether the punitive pattern is stronger among them.

1.2 What is new relative to *The Dark Side*?

The Dark Side study measured punitiveness and its correlates across the entire sample. This study keeps that core and adds:

- Items on the four goals of punishment, a seven-item retribution scale, and a forced ranking of the four goals that defines who counts as a retributivist.
- A revenge and proportionality section that separates revenge from just deserts and tests whether just deserts holds together as a construct.
- Items on social exclusion, essentialism about criminals, the belief that crime is a choice, and whether society owes anything to would-be offenders.
- Cognitive Reflection Test.
- A decision to kill a shark following a lethal shark attack
- Three crime vignettes for the sentencing task, with the vignette each participant saw recorded in the data.

1.3 Research questions

- Does punitiveness correlate with the same attitudes found in the Dark Side study, and is the correlation larger for the hostile attitudes than for fear of crime?
- Is ordinary support for retribution coherent, proportionate, and consistent with the Kantian limits, or does it line up with revenge?

- Are these patterns concentrated among the people who rank retribution first?

2. Hypotheses

Construct names point to the measures in Section 5. Where a prediction names a direction it is tested one-sided; the rest are tested two-sided.

2.1 Whole-sample replication of the Dark Side pattern

- H1.** Punitiveness correlates positively with each correlate in Section 5: due-process permissiveness, willingness to convict on uncertain evidence, hatred, anger, social exclusion, degradation, infliction of suffering, tolerance of prison violence, harsh prison conditions, revenge, perceived crime rates, and fear of crime.
- H2.** The hostile-aggression cluster (hatred, revenge, degradation, infliction of suffering, tolerance of prison violence) correlates more strongly with punitiveness than the crime-concerns cluster (perceived crime rates, fear of crime).
- H3.** The correlates are mostly positively correlated with one another, in line with a single punitiveness outlook, not a set of unrelated attitudes.

2.2 Construct validity of retribution (whole sample)

- H4.** Participants do not consistently endorse strict proportionality. Agreement with the proportionality item sits at or below the scale midpoint, and a sizeable share endorse punishment beyond the seriousness of the crime.
- H5.** Participants support punishment beyond what proportionality requires. Agreement that punishment should exceed the seriousness of the crime is higher than agreement that it should fall short of the harm caused.
- H6.** Just deserts does not hold together. Its two parts, that punishment is deserved and that it should be proportionate, correlate weakly and do not form a reliable scale.
- H7.** The Kantian limits (due process, reliable evidence, dispassion, dignity, freedom from needless suffering), scored as willingness to set the limit aside, do not line up with just deserts and are positively related to revenge.
- H8.** Willingness to set aside the Kantian limits correlates positively with punitiveness.
- H9.** Self-reported support for just deserts is higher than self-reported support for revenge.
- H10.** Revenge correlates more strongly with punitiveness than just deserts does.

2.3 Retributivists versus non-retributivists (confirmatory)

- H11.** The whole-sample relationships in H1 through H10 also hold within the retributivist group.
- H12.** A sizeable share of retributivists endorse restricting defendants' due-process rights, convicting on substandard evidence, hatred toward offenders, degradation of prisoners, infliction of suffering, tolerance of prison violence including sexual violence, harsh prison conditions, and inflicting pain on the shark; and each of these correlates with punitiveness.

- H13.** Retributivists score higher than non-retributivists on the hostile-aggression measures and the wider set of punitiveness correlates.
- H14.** Retributivists give longer prison sentences than non-retributivists in the crime vignette.
- H15.** Retributivists are more willing than non-retributivists to put the shark to death, and they endorse a more painful death.
- H16.** Retributivists hold stronger essentialist beliefs about where criminality comes from.
- H17.** Retributivists more strongly believe that crime is a choice.
- H18.** Retributivists side more with the prosecution than the defense.
- H19.** Retributivists more strongly reject the idea that society should care for would-be offenders.
- H20.** Retributivists are more willing than non-retributivists to set aside the limits on due process and reliable evidence.
- H21.** Retributivists score higher on punitiveness than non-retributivists.

2.4 Exploratory predictions

- E1.** Retributivists score higher than non-retributivists on Right-Wing Authoritarianism, Social Dominance Orientation, violence proneness, and racial resentment, and lower on cognitive reflection.
- E2.** Hostile solidarity (social exclusion of offenders [exclusion_1 to exclusion_3]) relates to punitiveness more strongly than Durkheimian solidarity (punishment as expressing shared norms and binding the law-abiding together [Solidarity_1 to Solidarity_3]).
- E3.** In their backward-looking explanations (the sentence justification [Sentence_OpenEnd1] and the goal-ranking explanation [Q333]), retributivists use more desert, proportionality, and censure language than non-retributivists.
- E4.** Within participants, principled-retributive language (desert, proportionality, censure) appears together with vengeful language (revenge, infliction of suffering) rather than in place of it. A positive correlation between the two across responses would show that ordinary just deserts is bound up with revenge.
- E5.** Desert language and proportionality language do not hold together in the open-ended responses, the text version of H6.
- E6.** Backward-looking explanations (why this sentence) are more retributive, and forward-looking explanations (what the sentence should accomplish) more consequentialist; the difference between them within a person indexes a retributive versus consequentialist motive.
- E7.** The language facets agree with the survey retributivism measures and add prediction of the sentence beyond the self-report scales.

E3 through E7 are exploratory and depend on the validation step in Section 6.7.

3. Design

3.1 Study type

An online survey with self-report measures. Everyone answers the same questions, with one exception: in the sentencing task each person reads one of three crime vignettes, assigned at random. The vignettes are included to increase the generalizability of the finding.

3.2 Randomization

The order of items within blocks is generally randomized, but the order of the blocks is fixed, with two exceptions. In the sentencing task a randomizer presents one of the three vignette blocks, with even assignment across participants. All three describe the same defendant, victim, and charge (second-degree murder) but differ in circumstance: a robbery that turns deadly, an episode of intimate-partner violence, and a killing inside a criminal operation. A field named Vignette records which one each participant saw (1, 2, or 3). The four personality blocks (RWA, SDO, violence proneness, racial resentment) appear in a random order to balance order effects; the Cognitive Reflection Test follows them in a fixed slot.

3.3 Order of presentation

The blocks run in this order:

Informed consent, Prolific ID, Introduction, Goals of punishment, Punitiveness, Revenge and proportionality, Exclusion and essentialism and choice, Due process, Crime concerns, Emotions, Harshness, Personality battery (RWA, SDO, violence proneness, racial resentment, in random order), Cognitive Reflection Test, Sentencing tasks (one of three vignettes), Shark scenario, Demographics, Feedback.

Two attention checks sit in the flow at fixed points: one in the punitiveness block (an instruction to pick a stated response) and one in the due-process block (a simple arithmetic item).

4. Sampling

4.1 Participants and recruitment

Participants, recruited on Prolific, form a representative sample of adults living in the United States along gender, race, ideology, and political leaning. They must be fluent in English, have an approval rating of at least 95 percent, and not have taken part in this study or its pilots.

4.2 Target sample size

We aim for 960 complete, valid responses after exclusions, double the 480 used in *Dark Side*. The larger number is needed because several hypotheses (H11 to H20) are tested only within the retributivist group, which will be a fraction of the sample, and because the dependent-correlation comparisons (H2, H10) and the group contrasts need power at the subgroup level. We recruit about 5 to 10 percent above target to cover exclusions.

4.3 Smallest detectable effects

With 960 participants and a two-sided test at $\alpha = .05$, the study has about 80 percent power to detect a correlation near $r = .09$. For a retributivist-versus-rest contrast with a split around 35 to 65 percent, it has about 80 percent power to detect a standardized mean difference near $d = 0.19$. These are the smallest effects the design can detect, not predictions about effect size.

4.4 Stopping rule

Data collection stops at the target number of valid responses. No hypotheses are tested before collection ends, and the target does not change in response to early results.

5. Measures and scoring

5.1 General conventions

Unless noted, attitude items use a seven-point agree-disagree scale scored 1 to 7. A few sections use seven-point scales with other anchors, for example not serious at all to very serious, noted where they appear. Each composite is the mean of its items after reverse coding, computed for anyone who answered at least half of the items. We report Cronbach's alpha and McDonald's omega for each. An item marked (R) is reverse coded, so that a higher score means more of the named construct. Names in brackets are the variable tags in the Qualtrics export.

5.2 Punitiveness

Punitiveness is the central measure and is built as in the *Dark Side* study. It combines support for more punishment [punishmore_1 (R), punishmore_2], rejection of parsimony [parsimony_1 (R), parsimony_2 (R)], support for three-strikes laws [threestrikes_1, threestrikes_2], support for life without parole [LWOP], support for the death penalty [deathpenalty], and the sentence given in the vignette [Sentence1], standardized within vignette (Section 5.11). We report two versions: a self-report scale from the first five parts, and a full index that adds the standardized sentence.

5.3 Goals of punishment and the definition of retributivism

The goals section has items for each justification: retribution [Retribution_1 to Retribution_7], deterrence [Deterrence_1 to Deterrence_3], incapacitation [Incapacitation_1, Incapacitation_2], rehabilitation [Rehabilitation_1, Rehabilitation_2], and solidarity [Solidarity_1 to Solidarity_3]. Each set is averaged into a subscale; the retribution subscale feeds the continuous measure of retributivism defined below.

Identifying retributivists is expected to be difficult, and that is why we use more than one measure. Three considerations lead us to expect that any single measure, and any self-report measure in particular, will be imperfect. First, demand characteristics: participants may report the goals they take to be expected or socially acceptable, which tends to favor the forward-looking, utilitarian justifications and to understate retribution. Second, intuitive retributivism: people often punish on the basis of desert, tracking the seriousness of the offense, while describing their own aims as utilitarian, and they are largely insensitive to the factors that a deterrence or incapacitation rationale should weigh, such as the likelihood of detection (Carlsmith, Darley, & Robinson, 2002). A participant can therefore be retributive in practice while reporting utilitarian motives. Third, limited introspective access: people may not consciously know why they want to punish, so a direct question about their goals need not recover the motive that is actually driving them. Each of these works against the self-report measures, which is why we add a behavioral measure (the sentencing ratio below) and an exploratory semantic measure (the retributive language of the open-ended responses, Section 6.7) that can register a retributive motive even when a participant does not report one.

Participants rank the four goals [Prioritization] and explain the ranking in their own words. We identify retributivists with three measures, plus a fourth that is exploratory. First, the ranking, which is the primary categorical definition: a participant who ranks "criminals deserve to be punished" first. Second, the relative retribution subscale, which is the primary continuous measure: retribution taken relative to the other three goals, that is, the retribution subscale with deterrence, incapacitation, and rehabilitation partialled out (equivalently, retribution minus the mean of the other three). This captures prioritizing

retribution rather than simply endorsing every goal, and it is the continuous measure used wherever retributivism is entered as a predictor. Third, the sentencing ratio (Section 5.11): when the forward-looking reasons for punishment are removed, the punishment a participant still imposes reflects desert, indexed as the share of the first sentence retained in the second (Sentence2 divided by Sentence1, for participants whose first sentence was above zero). A higher ratio, meaning a smaller reduction, indicates a more retributive motive; we use the ratio rather than the raw drop because the raw drop is bounded by the first sentence and is expected to pile up at zero, and we report the regression-adjusted version (the second sentence residualized on the first) as a robustness check. Fourth, and exploratory, the semantic measure: the retributive facets scored from the open-ended responses (Section 6.7), available only for participants who wrote substantive answers and used only after those facets are validated against the survey scales.

The ranking is the primary categorical grouping and the relative retribution subscale is the primary continuous measure. For the categorical contrasts (H11 to H21) we report a median split on the relative subscale, with the top and bottom thirds alongside it, and we report these contrasts again under the sentencing ratio, which may prove the sharpest of the measures. The sentencing ratio is used as a grouping or predictor variable for every dependent variable that is not itself the sentence, including hostile aggression (H13), the shark judgments (H15), essentialism (H16), the belief that crime is a choice (H17), the prosecution-defense leaning (H18), society's responsibility (H19), the due-process and evidence limits (H20), punitiveness (H21), and the personality measures (E1). It is not used for the sentence-based hypothesis H14, because it is built from the same sentence H14 tests. Wherever the sentencing ratio appears together with punitiveness, whether as a grouping variable, a predictor, an interaction term, or a correlate, we use the self-report punitiveness scale, which does not contain the sentence, rather than the full index, so the sentence cannot enter on both sides. The semantic measure is used the same way for non-sentence outcomes, with the added restriction that it is never used to group participants for the language hypotheses (E3 to E7), which are built from the same responses. We treat the measures as convergent: we report their pairwise agreement and the overlap among the groups they define, and we report every contrast under each measure rather than only the one that gives the strongest result.

For E2 the solidarity items are read two ways: a hostile reading (the exclusion items [exclusion_1 to exclusion_3]) and a Durkheimian reading (punishment as binding the law-abiding together and stating shared norms [Solidarity_2, Solidarity_3]); Solidarity_1 ("Removing criminals from society helps society cleanse itself") is dropped because it reads as exclusionary rather than Durkheimian. The exclusion items also serve as the social-exclusion correlate elsewhere, and we do not enter them twice in the same analysis.

5.4 Revenge, proportionality, and just deserts

Revenge is four items on society's right to get back at offenders [revenge_1 to revenge_4]. Proportionality is one direct item [proportionality] plus two items on whether punishment should exceed the seriousness of the crime [severity_1] or fall short of the harm [severity_2 (R)]. Just deserts is treated as two parts: that punishment is deserved (the purpose items, Retribution_1 to Retribution_4) and that it should be proportionate, scored from the proportionality item together with the two severity items reverse-keyed so that wanting punishment greater or less than the crime lowers proportionality [proportionality, severity_1 (R), severity_2 (R)]. H6 tests whether these two parts hold together; if they do not, we report that instead of forcing them into one score. For H9 we compare support for just deserts with support for revenge.

5.5 The Kantian limits

The limits that go with a defensible retributivism are measured in several places, scored as willingness to set the limit aside, so that a higher value reflects the less restrained, more punitive direction; items worded as protection or endorsement of a limit are reverse-coded accordingly. Due process is measured by permissiveness toward procedural violations [dueprocess_1,

dueprocess_2 (R), dueprocess_3]; reliable evidence by willingness to convict on weak or improper proof [uncertain_1 to uncertain_4]; dispassion by its opposite, the emotions in Section 5.7 (hatred and anger); dignity by support for degradation [degrad_1 (R), degrad_2, degrad_3]; and freedom from needless suffering by endorsement of suffering as a goal [suffer_1, suffer_2] and tolerance of prison violence [prisonvi_1 (R), prisonvi_2 (R)]. The degradation, suffering, and prison-violence items also enter the hostile-aggression composites in the same direction, so where the Kantian-limits measure and the hostile-aggression measure would appear in the same analysis we use only the non-overlapping limit components (due process, reliable evidence, and dispassion) to avoid double-counting. H7, H8, and H20 use these.

5.6 Crime concerns

Perceived crime rates [rates_1, rates_2] and fear of crime [fear_1 (R), fear_2, fear_3]. Together they make up the crime-concerns cluster that H2 contrasts with the hostile-aggression cluster.

5.7 Emotions toward offenders

Hatred [hate_1, hate_2 (R), hate_3] and anger [anger_1, anger_2]. The hatred scale belongs to the hostile-aggression cluster and also stands in, reversed, for dispassion among the Kantian limits.

5.8 Harshness toward offenders

Degradation [degrad_1 (R), degrad_2, degrad_3], infliction of suffering [suffer_1, suffer_2], tolerance of prison violence including sexual violence [prisonvi_1 (R), prisonvi_2 (R)], and support for harsh prison conditions [harsh_1, harsh_2 (R), harsh_3]. Degradation, suffering, and prison violence belong to the hostile-aggression cluster.

5.9 Exclusion, essentialism, choice, and society's responsibility

Social exclusion of offenders [exclusion_1 to exclusion_3], essentialism about criminals [essential_1, essential_2, essential_3 (R), essential_4], the belief that crime is a choice [choice_1, choice_2], and rejection of society's responsibility for offenders [Society_1 (R), Society_2 (R), Society_3 (R)]. Exclusion, essentialism, and choice support H16 and H17; the society items, reversed, support H19.

5.10 The shark scenario

A short scenario asks whether a shark that killed a child should be put to death [Shark_1, yes or no], how confident the participant is [Shark_2], and, if it is killed, how painful the death should be [Shark_3]. A shark cannot be deterred or reformed and, given that it lacks human agency, defies a central tenet of Kantian retributivism. H12 and H15 use these items.

5.11 The sentencing vignette

After the vignette, participants set a sentence from 0 to 50 years, where 50 stands for life [Sentence1]. They then read a set of added facts that take away the forward-looking reasons for punishment (the defendant is permanently disabled and no longer a danger, and legal procedural rules remove any prospect of general deterrence) and set a second sentence [Sentence2]. Because the three vignettes differ in severity, the first sentence is standardized within vignette before it is pooled or added to the punitiveness index. The punishment a participant still imposes once those reasons are gone reflects desert; we index this as the share of the first sentence retained in the second, Sentence2 divided by Sentence1, computed for participants whose first sentence was above zero. A higher ratio, meaning a smaller reduction,

indicates a more retributive motive. Because the ratio is a proportion it is comparable across vignettes, and we confirm that results hold within each vignette; we also report the regression-adjusted version (the second sentence residualized on the first) as a robustness check. This index is one of the ways we identify retributivists (Section 5.3): it serves as a grouping or predictor variable for every outcome except the sentence-based hypothesis H14, which it cannot define without circularity because it is built from the sentence that hypothesis tests, and it is also analyzed as an outcome in its own right. Wherever it is examined together with punitiveness, we use the sentence-free punitiveness scale so the first sentence cannot enter on both sides.

5.12 Personality

Right-Wing Authoritarianism, five items taken from Bizumic & Duckitt (2018), adapted from Altemeyer [rwa_1 to rwa_5, rwa_3 (R)]. Social Dominance Orientation, the eight-item SDO7 short form (Ho et al., 2015) [sdo_1 to sdo_8, with sdo_3, sdo_4, sdo_7, and sdo_8 reversed]. Violence proneness, four items on endorsing violence to defend property, in child-rearing, against suspected terrorists, and in war [vprone_1 to vprone_4]. Racial resentment, the four standard items (Kinder & Sanders, 1996) [Raceresent_1 (R), Raceresent_2, Raceresent_3, Raceresent_4 (R)]. Cognitive Reflection Test, the three standard items [CRT_1, CRT_2, CRT_3], scored as the number correct from 0 to 3, with higher scores meaning more reflective answers.

5.13 Demographics and orientation

Age [age], gender [gender], race and ethnicity [race], education [education], a single liberal-conservative item [lib/conserv], party identification [PoliticalID], and two items on whether the participant sides more with the prosecution or the defense [leanPros_1, leanPros_2], used for H18. Political orientation and party identification also serve as covariates.

5.14 Attention checks

Two checks catch inattentive responding: an instructed-response item in the punitiveness block [AttnCheck1] and an arithmetic item in the due-process block [AttnCheck2, where 8 plus 7 is 15]. Section 6.2 gives their role in exclusions.

5.15 Open-ended responses and the retributive facets

The study collects three open-ended responses: the sentence justification [Sentence_OpenEnd1], a statement of what the sentence should accomplish [Sentence_OpenEnd2], and the explanation of the goal ranking [Q333]. We analyze all three as text, and we score them separately, not as one combined passage. The justification and the ranking explanation look back at what the offender did, where desert and proportionality language tends to appear, while the accomplishment item looks ahead, where deterrence, incapacitation, and rehabilitation language tends to appear. Combining them would mix these two kinds of reasoning, which is the distinction the study is trying to keep. All three items use a soft prompt that asks for an answer but lets the participant continue without one, so we encourage substantive responses without forcing them.

Each response is scored on a set of retributive facets instead of a single prosocial-versus-dark contrast. Four facets stand for principled retribution: desert (the offender earned the punishment), proportionality (a punishment that fits the crime and no more), censure (punishment as a statement that the act was wrong), and moral balance (paying a debt or restoring order). Two stand for its harder edge: revenge (getting even) and suffering for its own sake. The three consequentialist goals (deterrence, incapacitation, rehabilitation)

are scored as well, so retributive language can be compared against them. More than one facet can apply to a response, since a single answer can show several at once, and this is needed to test whether the principled and vengeful facets appear together. The facets line up with the closed-ended items (just desert purpose items, Retribution 1-4, proportionality with the proportionality item, revenge with the revenge items, suffering with the suffer items), which lets the two kinds of measure check each other.

6. Analysis plan

6.1 General approach

Each composite is the mean of its items after reverse coding, with a reliability coefficient reported alongside it. The order of work follows the *Dark Side* study: build the composites, check their reliabilities, standardize the sentence within vignette, run the confirmatory tests, then the exploratory ones. Associations are tested with Pearson correlations and 95 percent confidence intervals. When a set of related correlations is tested together, as in H1, we control the false discovery rate with the Benjamini-Hochberg procedure and report both adjusted and unadjusted p-values. The closed-ended analyses follow the statistical workflow used in the Dark Side study; the open-ended responses are scored in a separate text-processing step described in Section 6.7. The analysis code and materials will be posted with the data.

6.2 Data cleaning and exclusions

We exclude participants who fail either attention check, who do not finish the survey, or who fall short of the Prolific and residency criteria in Section 4.1. Failing one check is enough, since both are easy for an attentive respondent. We also examine completion time and long strings of identical answers and report them, but neither on its own removes a participant from the confirmatory analyses. Exclusions are applied before any hypothesis test, and we report how many drop out at each step. We set no separate outlier rule for the attitude composites; the sentence is bounded at 0 and 50 by the slider, so we do not trim it.

6.3 Whole-sample replication (H1 to H3)

H1 is the set of correlations between punitiveness and each correlate, with the false discovery rate controlled across the set. H2 compares two overlapping correlations from the same people, punitiveness with the hostile-aggression cluster against punitiveness with the crime-concerns cluster, using Steiger's test. H3 looks at the correlations among the correlates, summarized by the share that are positive and by the first eigenvalue and scree of an exploratory factor analysis, to see whether one punitiveness dimension runs through them.

6.4 Construct validity of retribution (H4 to H10)

H4 and H5 compare item and composite means against the scale midpoint with one-sample tests and report the share of participants who endorse punishment beyond proportionality. H6 tests whether just deserts holds together, using the correlation between the purpose composite (Retribution_1 to Retribution_4) and the proportionality component, the alpha of the two together, and a confirmatory factor model in which poor fit or a near-zero loading counts as failure to cohere. H7 correlates the limit measures with revenge and with just deserts, expecting a negative relation to revenge (the limits set aside) but none to just deserts. H8 correlates the limit measures with punitiveness, with a negative relation expected. H9 compares mean support for just deserts with mean support for revenge using a paired test. H10 compares punitiveness-with-revenge against punitiveness-with-just-deserts, again with Steiger's test.

6.5 Retributivist contrasts (H11 to H20)

Retributivists are defined as in Section 5.3. H11 reruns the H1 to H10 tests inside the retributivist group. H12 reports, for each listed attitude, the share of retributivists who endorse it (agreement above the midpoint) with confidence intervals, and its correlation with punitiveness. H13, H14, H16, H17, H18, and H19 compare retributivists with everyone else using Welch t-tests and Cohen's d, with the standardized sentence used for H14. H15 compares the shark judgments between groups, with a logistic regression for the yes-or-no decision and a t-test for the pain rating. We also run every group contrast with retributivism entered continuously, as a regression on the retribution subscale, so the result does not depend on the split.

6.6 Exploratory analyses

E1 compares the personality measures between groups with the same methods. E2 compares punitiveness-with-hostile-solidarity against punitiveness-with-Durkheimian-solidarity using Steiger's test. The language analysis (E3 to E7) is set out in Section 6.7. We also report the factor structure of the correlates, the change from the first to the second sentence as a function of retributivism, and political orientation and the prosecution-defense leaning as moderators. These analyses are exploratory and are not corrected as a confirmatory family.

6.7 Language analysis of the open-ended responses

The language analysis measures retributive reasoning from the three open-ended items in Section 5.15. It uses the text-processing setup from the Dark Side study, but with the categories changed from a prosocial-versus-dark split to the retributive facets, because the question here is how much retributive reasoning a response contains and of what kind, not whether it sounds prosocial.

Coding is done by an ensemble of language models in place of human raters, at least three of them from different developers and including one open-weight model. Each model scores every response on the facets, allowing more than one facet per response, and makes the single principled-versus-vengeful judgment. The facet score for a response is the average across models, and the holistic judgment is the majority call. We report agreement across the models for each facet, an intraclass correlation for the 0-to-1 scores and Krippendorff's alpha for the categorical judgment, which plays the part that inter-rater reliability plays with human coders. The models are run at a fixed setting and pinned versions, with all raw output saved so the scoring can be repeated.

Agreement among the models is reliability, not validity. Models trained on similar data can share the same bias, and the line between principled desert and revenge is the thing this study examines, so we do not treat the ensemble as ground truth. Two safeguards address this. First, two measures that do not use a language model are scored alongside the ensemble: a word list, whose retributive entries are wider than the Dark Side list and whose known false matches are removed, so the adverb "just" no longer counts toward desert and no word counts toward two opposing facets; and similarity to short reference passages for each facet, computed with more than one embedding model. Because the word list does not use a language model, its agreement with the ensemble is a more independent check than the models' agreement with one another. Second, the facet scores are validated against the closed-ended scales, which are external to the text: the desert score should track the desert item and the retribution composite, revenge language the revenge scale, suffering language the suffer items, and proportionality language the proportionality item, and each facet should relate more strongly to its own scale than to an unrelated one. A facet supports a claim only if the models agree at an intraclass correlation of at least .60 and the facet correlates at least .30 with its matching

survey scale. We also report whether revenge and suffering language predict the shark pain rating and the sentence.

The language hypotheses are tested as follows. For E3 we compare facet scores between retributivists and others, and regress them on the retribution subscale, using the backward-looking texts. For E4 we estimate the correlation across responses between the principled facets (desert, proportionality, censure) and the vengeful facets (revenge, suffering). Because both sides come from the same automated scoring, one method's tendency to assign several facets at once could create this correlation on its own, so we require the same co-occurrence to show up in the non-model word-list measure, and we compare it with the relationship between the desert and revenge scales in the survey, before reading it as evidence that ordinary just deserts is bound up with revenge. For E5 we test whether desert and proportionality language hold together, the text version of H6. For E6 we compare each person's facets in the backward-looking item with the forward-looking item. For E7 we add the facet scores to a model predicting the within-vignette sentence on top of the survey measures, using the eight-item punitiveness scale rather than the full index, because the full index already contains the sentence.

E3 through E7 form one exploratory family; we control the false discovery rate within it and report effect sizes and confidence intervals. Word count is a covariate, since people differ in how much they write. As in the *Dark Side* study, the absolute level of the similarity scores shifts across embedding models while the ordering of responses stays stable, and we report that for the facet scores as well. The model set and versions, the scoring prompts, the word lists, the raw model output, and the analysis code go on the project's OSF page.

6.8 Inference criteria

The cutoff for the confirmatory tests is $\alpha = .05$. Predictions that name a direction are tested one-sided in that direction; the rest are two-sided. Every test reports an effect size and a confidence interval, and the families named above use corrected p-values.

7. Prior knowledge of the data

No data have been collected for this study yet. The instrument and the plan draw on the Dark Side of Punishment, which shape the predictions but contain none of the present data.

References

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Appendix A. Measure summary

Items marked (R) are reverse coded. Composite scores are item means after reverse coding. Scales are seven-point unless noted.

Construct	Items (export tags)	Scale	Reverse-keyed	Scoring / role
Punitiveness (self-report)	punishmore_1-2, parsimony_1-2, threestrikes_1-2, LWOP, deathpenalty	7-pt	punishmore_1, parsimony_1-2	Mean; central measure
Punitiveness (full index)	self-report scale + Sentence1	mixed	-	Adds z-scored sentence
Retribution tenets	Retribution_1-7	7-pt	none	Continuous retributivism (main predictor)
Deterrence	Deterrence_1-3	7-pt	none	Mean subscale
Incapacitation	Incapacitation_1-2	7-pt	none	Mean subscale
Rehabilitation	Rehabilitation_1-2	7-pt	none	Mean subscale
Solidarity	Solidarity_1-2	7-pt	none	Durkheimian solidarity (E2)
Goal ranking	Prioritization; Q333 (open)	rank	-	Categorical retributivist = retribution ranked 1st
Revenge	revenge_1-4	7-pt	none	Mean; compared with just deserts
Proportionality	proportionality, severity_1, severity_2	7-pt	severity_2	H4, H5
Just deserts	Retribution_4 + proportionality	7-pt	-	Coherence tested, not assumed (H6)
Due-process permissiveness	dueprocess_1-3	7-pt	dueprocess_2	Kantian limit (set aside)
Uncertain evidence	uncertain_1-4	7-pt	none	Kantian limit (set aside)
Hatred	hate_1-3	7-pt	hate_2	Hostile-aggression; dispassion (rev.)
Anger	anger_1-2	7-pt	none	Emotions
Degradation	degrad_1-3	7-pt	degrad_1	Hostile-aggression; dignity
Infliction of suffering	suffer_1-2	7-pt	none	Hostile-aggression
Prison violence tolerance	prisonvi_1-2	7-pt	prisonvi_1, prisonvi_2	Hostile-aggression
Harsh conditions	harsh_1-3	7-pt	harsh_2	Harshness
Perceived crime rates	rates_1-2	7-pt	none	Crime-concerns cluster
Fear of crime	fear_1-3	7-pt	fear_1	Crime-concerns cluster
Social exclusion	exclusion_1-3	7-pt	none	Hostile solidarity (E2); exclusion section
Essentialism	essential_1-4	7-pt	essential_3	H16
Crime as choice	choice_1-2	7-pt	none	H17

Society's responsibility (reject)	Society_1-3	7-pt	Society_1-3	H19 (reversed)
Sentence (vignette)	Sentence1, Sentence2; open-ends	0-50 yrs	-	z within vignette; H14
Shark	Shark_1 (y/n), Shark_2, Shark_3	mixed	none	H12, H15
RWA	rwa_1-5	7-pt	rwa_3	E1
SDO7 (short)	sdo_1-8	7-pt	sdo_3, sdo_4, sdo_7, sdo_8	E1
Violence proneness	vprone_1-4	7-pt	none	E1
Racial resentment	Raceresent_1-4	7-pt	Raceresent_1, Raceresent_4	E1
Cognitive reflection	CRT_1-3	numeric	-	Number correct (0-3); E1
Demographics	age, gender, race, education, lib/conserv, PoliticalID, leanPros_1-2	mixed	-	Covariates; H18
Attention checks	AttnCheck1, AttnCheck2	-	-	Exclusion screen

Appendix B. Full item wording

Exact wording from the Qualtrics instrument. Items marked (R) are reverse coded. Instruction and display-only screens are left out. Response anchors appear where a block begins.

Goals of punishment

Response scale: Strongly disagree / Neutral / Strongly agree

Retribution_1. Society has a moral right to punish criminals

Retribution_2. Justice requires that crimes be punished

Retribution_3. Justice is restored when criminals are punished

Retribution_4. Punishment is justified because criminals deserve to be punished

Retribution_5. The state has a duty to punish its criminals

Response scale: strongly disagree / neutral / strongly agree

Retribution_6. Criminal punishment is a good way to get criminals to repay their debt to society

Retribution_7. Punishment negates the unfair advantage that criminals gain over law-abiding citizens

Response scale: Strongly disagree / Neutral / Strongly agree

Deterrence_1. Punishment is justified because it deters others from committing crimes

Deterrence_2. Punishment makes it clear that crime does not pay

Deterrence_3. Without a serious threat of punishment, crime rates would skyrocket

Incapacitation_1. Society is safest when criminals are locked behind bars

Incapacitation_2. Imprisonment is justified because it removes criminals off the streets

Rehabilitation_1. Imprisonment is justified because it prepares prisoners for their eventual release back into society

Rehabilitation_2. Prison programs can make prisoners become productive members of society

Solidarity_1. Removing criminals from society helps society cleanse itself

Solidarity_2. Punishing criminals brings the law-abiding society closer together

Solidarity_3. Punishing criminals expresses society's norms

Response scale: Criminals deserve to be punished / Punishment deters others from committing crimes / While they are locked up, prisoners cannot commit other crimes / Prisoners get rehabilitated in prison

Prioritization. Please read the list of four reasons that are often used to justify criminal punishment: Criminals deserve to be punished Punishment deters others from committing crimes While they are locked up, prisoners cannot commit other crimes Prisoners get rehabilitated in prison Please rank how you prioritize these reasons in descending order (rank the best reason first and the weakest reason last). My ranked order of reasons for punishment is --

Q333. Please explain why you ranked the reasons for punishment in that order

Punitiveness

Response scale: Strongly disagree / Neutral / Strongly agree

punishmore_1 (R). The current level of punishment in the United States is too high.

punishmore_2. The solution to crime is more punishment.

parsimony_1 (R). Criminal punishments should be as light as possible.

parsimony_2 (R). The state should not punish more than is absolutely necessary.

threestrikes_1. People convicted three times for a felony should be imprisoned for life.

threestrikes_2. People convicted three times for shoplifting should serve at least 25 years in prison.

LWOP. Convicted murderers should never be allowed out of prison.

Response scale: Strongly oppose / Neutral / Strongly support

deathpenalty. How much do you support or oppose the use of the death penalty to punish people convicted of first-degree murder?

Response scale: Strongly oppose / Neutral / Strongly agree

AttnCheck1. To make sure that the experiment is running correctly, please select the “strongly agree” option.

Revenge and proportionality

Response scale: Strongly disagree / Neutral / Strongly agree

revenge_1. Society has the right to take revenge on criminal offenders.

revenge_2. Society should punish to get back at criminal offenders.

revenge_3. A father should not be prosecuted for killing the man who sexually assaulted his 9-year old daughter.

revenge_4. Rather than showing criminals so much tolerance, we should be punishing them more severely.

proportionality. Punishment should be proportionate to the severity of the crime, no more and no less.

severity_1. Punishment should be greater than the seriousness of the crime.

severity_2 (R). Punishment should be less severe than the harmfulness of the crime.

Exclusion, essentialism, choice, and society

Response scale: Strongly disagree / Neutral / Strongly agree

exclusion_1. By breaking society’s laws, criminals exclude themselves from membership in society.

exclusion_2. Having chosen to break the law, criminals do not deserve to enjoy the benefits that society provides its members.

exclusion_3. Punishment communicates to criminals that they have no place in society.

essential_1. Once a criminal, always a criminal.

essential_2. The people who commit crimes do so because criminal behavior is in their nature.

essential_3 (R). The people who commit crimes do so because of their tough life circumstances.

essential_4. Criminals have only themselves to blame for their punishment.

Society_1 (R). When a person breaks the law, is not just a personal failure but also society’s failure

Society_2 (R). The state should take better care of people so that they will choose not to commit crimes

Society_3 (R). Society should embrace all of its members, including those who break its laws

choice_1. People sitting in prison are there because they chose to live a life of crime.

choice_2. Committing a crime is always a choice.

Due process and evidence

Response scale: 12 / 16 / 17 / 15 / 14 / 13 / 18

AttnCheck2. What is 8+7 equal to?

Response scale: Strongly disagree / Neutral / Strongly agree

dueprocess_1. The legal system’s loopholes and technicalities makes it too difficult to convict criminals for their crimes.

dueprocess_2 (R). No matter how abhorrent the crime is, every defendant deserves the protection of due process of law.

dueprocess_3. People charged with committing violent crimes should be convicted even if police investigators cut some corners.

uncertain_1. Jailhouse informants can be trusted when they testify that a cell-mate had confessed to committing a crime.

uncertain_2. Generally, the police make an arrest only when they are sure who committed the crime.

uncertain_3. When it is the suspect’s word against the police officer’s word, we should believe the police.

uncertain_4. If police officers know that a defendant is guilty, it is okay for them to lie when testifying in court to help obtain a conviction.

Crime concerns

Response scale: Not serious at all / Very serious

Q315. Please respond to the following questions

rates_1. How serious is the problem of crime in the United States?

Response scale: 1 - Not at all / 2 / 3 / 4 / 5 / 6 / 7 - Very much so

rates_2. Do you feel that violent crime just keeps Rising?

Response scale: Not safe at all / Very safe

fear_1 (R). How safe do you feel walking at night in the vicinity of your home?

Response scale: Not at all / Very much so

fear_2. How much do you fear that you will be a victim of a serious crime?

fear_3. How much do you fear that someone will break into your home in the next 12 months?

Emotions

Response scale: Not at all / Very much so

hate_1. I feel hatred towards criminals.

hate_2 (R). I am sympathetic towards criminals.

hate_3. I feel that criminals are detestable human beings.

anger_1. I feel anger towards criminals.

anger_2. The crime problem makes me feel angry.

Harshness

Response scale: Strongly disagree / Neutral / Strongly agree

degrad_1 (R). Regardless of their wrongdoing, prisoners deserve to be treated with dignity and respect.

degrad_2. Punishment communicates to criminals that society has low regard for them.

degrad_3. Punishment should humiliate the criminal.

suffer_1. Infliction of suffering is a permissible goal of criminal punishment.

suffer_2. Punishment without suffering is not punishment.

prisonvi_1 (R). I feel bad when a prisoner is assaulted in prison.

prisonvi_2 (R). Sexual assault of prisoners is a serious societal problem.

harsh_1. The conditions in prisons should be extremely harsh.

harsh_2 (R). Prisoners deserve decent living conditions.

harsh_3. Our prisons are too comfortable.

Right-Wing Authoritarianism

Response scale: Strongly disagree / Neutral / Strongly agree

rwa_1. What our country needs most is discipline, with everyone following our leaders in unity.

rwa_2. God's laws about abortion, pornography, and marriage must be strictly followed before it is too late.

rwa_3 (R). There is nothing wrong with premarital sexual intercourse.

rwa_4. Our society needs tougher government and stricter laws.

rwa_5. To preserve law and order, we should crack down harder on troublemakers.

Social Dominance Orientation

Response scale: Strongly disagree / Neutral / Strongly agree

sdo_1. An ideal society requires some groups to be on top and others to be on the bottom.

sdo_2. Some groups of people are simply inferior to other groups.

sdo_3 (R). No one group should dominate in society.

sdo_4 (R). Groups at the bottom of the social order are just as deserving as groups at the top.

sdo_5. Group equality should not be our primary goal.

sdo_6. It is unjust to try to make groups equal.

sdo_7 (R). We should do what we can to equalize conditions for different social groups.

sdo_8 (R). We should work to give all groups an equal chance to succeed.

Violence proneness

Response scale: Strongly disagree / Neutral / Strongly agree

vprone_1. A person has the right to kill to defend their property.

vprone_2. Physical punishment is necessary for raising children.

vprone_3. There is nothing wrong in torturing terrorists to obtain information about the threat they may pose to society.

vprone_4. Imagine that the United States is at war with a foreign country. After months of heavy losses, the US ground invasion has ground to a halt, with experts estimating an additional 20,000 US military fatalities. One suggestion to end the war is to drop a nuclear weapon on the enemy's capitol, which could kill up to 200,000 civilians. To what extent do you agree with dropping the nuclear weapon on the enemy's capitol?

Racial resentment

Response scale: Strongly disagree / Neutral / Strongly agree

Raceresent_1 (R). Generations of slavery and discrimination have created conditions that make it difficult for blacks to work their way out of the lower class.

Raceresent_2. It's really a matter of some people not trying hard enough; if Blacks would only try harder they could be as well off as Whites.

Raceresent_3. Irish, Italian, Jewish, and many other minorities overcame prejudice and worked their way up. Blacks should do the same without any special favors.

Raceresent_4 (R). Over the past few years, Blacks have gotten less than they deserve.

Cognitive Reflection Test

Response scale: cents

CRT_1. A bat and a ball cost \$1.10 in total. The bat costs a dollar more than the ball. How much does the ball cost? The ball costs:

Response scale: Minutes

CRT_2. If it takes 5 machines 5 minutes to make 5 widgets, how long would it take 100 machines to make 100 widgets? It would take:

CRT_3. In a lake, there is a patch of lily pads. Every day, the patch doubles in size. If it takes 48 days for the patch to cover the entire lake, how long would it take for the patch to cover half of the lake? Covering half of the lake would take:

Sentencing (vignette 1 shown; vignettes 2 and 3 parallel)

Response scale: Years

Sentence1_V1. What sentence would you give Darryl Smith for killing Frankie Williams? I would sentence Darryl to this number of years in prison: (for a life sentence, mark 50 years)

Sentence2_V1. What sentence would you give Darryl under these conditions? I would sentence Darryl to this number of years in prison: (for a life sentence, mark 50 years)

Sentence_OpenEnd1_V1. Please explain why you recommended this sentence for Darryl.

Sentence_OpenEnd2_V1. What do you hope this sentence will accomplish?

Shark scenario

Response scale: No, I do not support putting it to death / Yes, I support putting it to death

Shark_1. At dusk one evening, Melissa, a 10-year old star gymnast, was swimming with her family at a popular beach. The beach was known to be very safe for swimmers even though it was fairly close to a breeding grounds for a particular breed of sharks. Without warning, Melissa was attacked by a large shark and was severely bitten in her abdomen and neck. Within minutes of the attack, Melissa was dead. Two days later, local fishermen found a live shark whose fin was tangled in their net. The shark was believed to be of the same species as the one that killed Melissa. The authorities must now decide what to do with the captured shark. Do you support putting the shark to death?

Response scale: Weakly confident / Highly confident

Shark_2. How confident are you in your response?

Response scale: Not painful at all / Very painful

Shark_3. Assuming that the authorities decided to put the shark to death, how painful should the death be?

Demographics and orientation

Response scale: Years

age. What is your age?

Response scale: Male / Female / Other

gender. What is your gender?

Response scale: White, non-Hispanic / Black / African-American / Asian / Hispanic / Latino / Middle Eastern / Native American / Pacific Islander

race. What is your race/ethnicity?

Response scale: Did not graduate high school / High school education only / 1-2 years post high school / 3-4 years post high school / 5-6 years post high school / More than 6 years post high school

education. What is your level of education?

Response scale: Strongly liberal / Strongly conservative

lib/conserv. Do you describe yourself as more liberal or more conservative?

Response scale: Democrat / Republican / Independent / other / I do not identify myself politically

PoliticalID. How do you describe yourself politically? I identify mostly as a:

Response scale: The prosecutor / The defense attorney / About the same

leanPros_1. When you think of a criminal trial in real life, with whom do you identify more? I identify more with (mark one):

Response scale: Prosecutor / Defense attorney / About the same

leanPros_2. Imagine that you worked as a lawyer in the criminal justice system. Which role would you rather serve? I would rather work as a (mark one):